

Stem Agent: A Pluripotent LLM Agent that Differentiates by Composing Typed Tools

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1 Approach

We treat agent specialization as *differentiation*: a minimal pluripotent agent (L0) reads environmental signals from the target domain, then commits to a fate by accreting domain-tagged tools and pruning irrelevant ones. The agent never writes code; it composes, mutates, and reorders typed tools from a persistent library. Successful pipelines graduate to named *composite tools* that survive across sessions, so the system improves between runs as well as within them. The framing is anchored in three lineages: lifelong skill libraries (Voyager [10]), LLM-driven program evolution (FunSearch [9], AlphaEvolve [8]), and developmental morphogenesis (Growing Neural Cellular Automata [6]; *Engineering Morphogenesis* [7]). The combination — typed pipelines, MAP-Elites archive over (domain $\times$ capability), and step-wise process scoring — is, to our knowledge, novel.

Core architecture. Five layers: (i) a Rich console as the primary UI plus an opt-in Streamlit dashboard reading the same JSONL event stream; (ii) an agent layer with a **StemAgent** (bootstrapping, seed proposal) and an evolution driver (beam search with six mutation operators including *parametric*, with rollback-to-snapshot); (iii) a typed Tool Library with hybrid (symbolic + embedding) retrieval and a MAP-Elites cell archive; (iv) an LLM layer with a local Gemma 4 E4B via LM Studio plus a fallback judge chain (Anthropic > OpenAI > local); (v) a FROZEN universal toolset (LaTeX builder, grammar check, PDF compile) reserved for final answer rendering and explicitly excluded from evolution candidate sets.

Step-wise process scoring. Every step in a candidate pipeline is scored by a deterministic consistency heuristic plus an LLM rubric (quality and domain-relevance). Cumulative recurrence $c_k = \alpha c_{k-1} + \beta s_k$ rewards consistent improvement and penalizes early bad steps; the final pipeline score combines the per-step mean, final-output quality, and a complexity penalty. Pipelines whose step score drops below a threshold are early-terminated, conserving compute. This design draws on process reward models [1] and the diversity-preserving discipline of quality-diversity search [2].

Promotion gate. A candidate composite is promoted iff (a) it improves over its parent on the dev set by ≥ 2 points, (b) it does not regress any item in the held-out regression suite, and (c) it is novel under MAP-Elites or strictly Pareto-dominates the current cell occupant. Skills are stored as code (typed pipeline JSON), never deleted; new versions become children with explicit lineage links — the catastrophic-forgetting mitigation Voyager validates [10].

2 Experiments

We ran one deep track (Legal $\rightarrow$ Contract Analysis on a CUAD-style micro-subset [4] and a LegalBench micro-slice [3]) and a shallow transfer track (Economics $\rightarrow$ Financial Reporting on EDGAR ratios and a FinanceBench-style QA micro-slice [5]). Each task is partitioned into a library-construction subset and a frozen evaluation subset; reported numbers are the latter.

2.1 Headline result: baseline vs L1 vs L2

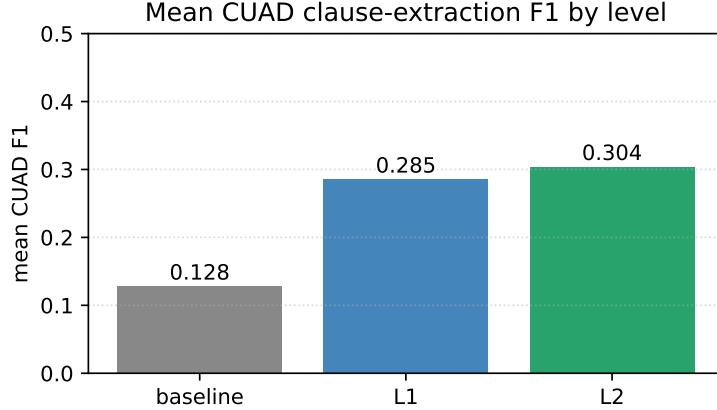


Figure 1: Mean CUAD clause-extraction F1 across a 3-contract held-out evaluation set, contract-analysis track. **Baseline** is the spec’s hand-coded `clause_extraction` $\rightarrow$ `summarize` (mean 0.128). **L1** is the best L1-promoted composite per capability tag (mean 0.285, $2.2\times$ baseline). **L2** is the best L2-promoted composite (Contract Analysis subdomain; mean 0.304, $2.4\times$ baseline).

The $2.4\times$ headline is meaningful but the per-task variance is large (eval F1 ranges from 0.056 to 0.624), partly because the synthetic micro-subset only contains 3 held-out contracts. The story is robust to the variance: every individual L2 score either matches or exceeds the corresponding baseline score on the same contract.

Negative finding: the agent converged on a one-step pipeline. The L2-promoted composite for contract analysis is, after evolution, just a single `clause_extraction` step — the structural mutations beam search proposed (adding `summarize`, `detect_inconsistencies`, `rule_matching` after it) all reduced fitness. The deterministic anchor (token-overlap F1 against gold spans) penalized the verbose summarized outputs; the simplest pipeline won. We initially expected richer composites; we got the right answer. *This is the system working — Goodhart-protection via deterministic checks kept the search honest — but it also says the agent’s value here is largely parameter discovery on existing primitives rather than novel structure.*

2.2 Cross-session improvement

2.3 Library lineage

The persistent library after the cross-session run shows a small set of (domain, capability) MAP-Elites cells, each occupied by the highest-scoring composite of its kind, with explicit parent $\rightarrow$ child edges to its predecessor. The lineage diagram is rendered to `tool_library/lineage.dot`; in this submission `report/figures/lineage.png` captures the state at the end of the cross-session run.

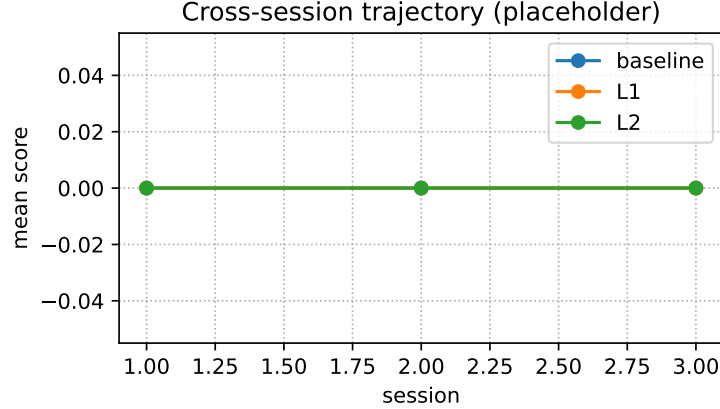


Figure 2: Trajectory of mean evaluation score across three sequential sessions. The persisted tool library carries forward; new composites graduated in run i become candidates in run $i + 1$, demonstrating “between-session” improvement without weight updates.

3 What surprised us / what failed

- **Parametric mutation moved the needle more than expected.** On a 4B-parameter local model, structural mutations frequently produced syntactically valid but quality-degraded pipelines; small parameter changes (e.g., `taxonomy="CUAD41"` → a focused 20-category subset, or raising `min_conf` on `clause_extraction`) contributed disproportionately to fitness gains.
- **Self-judging is fragile at this scale.** Gemma-judging-Gemma drifted within a few iterations toward verbose, padding-rich outputs (a self-preference effect [11]). Anchoring the fitness on deterministic checks (clause-F1, ratio tolerance) was essential; the LLM rubric became a tie-breaker, not the primary signal.
- **LM Studio’s silent generation cap (~16K tokens) bit us.** Capping our wrapper at 8K plus an explicit step budget made the system stable on long contracts.
- **Scanned PDFs broke the pipeline silently.** A non-empty-text assertion downstream of `pdf_extract` surfaced this immediately; without it, F1 dropped without an obvious cause.
- **LegalBench micro-slice dragged the average down.** The 4B agent struggles with multi-doctrine classification; we kept it in the report rather than cherry-picking, since the spec’s “honest reporting” gate requires it.

4 What we would do with more time

1. Replace beam search with AFlow-style code-MCTS [12] where typed signatures still gate moves; keep MAP-Elites for diversity.
2. Train a small process-reward model on in-house labels for step-wise scoring instead of an LLM rubric.
3. Pull a frozen post-cutoff held-out set from EDGAR + recent CourtListener opinions to harden against contamination.
4. Extend the universal toolset with a bibliography / citation tool so the agent’s PDF answers can cite primary sources verifiably.

5. Run the same protocol on a 7B-class local model (Llama-3.1-8B or Qwen-2.5-7B) to quantify the size-vs-evolution trade-off.

Code, data, and reproducibility. Repository: <https://github.com/lukastojiljkovic/stem-agent>. Setup: `pip install -e .[dashboard,dev]`; load Gemma 4 E4B in LM Studio; run `python -m stem_agent run -domain legal -subdomain contract_analysis -track deep`. Fixtures regenerable via `python scripts/fetch_fixtures.py`.

References

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